

RESEARCH ARTICLE

Decoupled Parallel Attention with Global Coupling for Infrared and Visible Image Fusion

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ABSTRACT

Infrared and visible imaging sensors are widely deployed on satellites and unmanned aerial vehicles for Earth observation, yet relying on a single modality fails to deliver reliable scene interpretation in complex environments. Infrared imagery highlights thermal targets but lacks texture, whereas visible imagery preserves background detail but degrades at night or under low illumination. Fusing the two modalities is therefore a key enabler of all-weather, all-day remote sensing image interpretation. Existing fusion approaches rely heavily on empirically designed fusion rules, and convolution-based networks capture local features well but model global context poorly, which leads to the loss of small targets and fine structures. This paper proposes an end-to-end fusion network that couples a multi-attention residual module (MARM) with a Swin Transformer-based module (STM). MARM cascades a coordinate attention block with a residual block and parallel spatial and channel attention blocks, so that thermally salient regions and texture-rich areas are emphasized while redundant responses are suppressed. STM is built upon the Swin Transformer and employs window-based and shifted-window multi-head self-attention to model long-range dependencies at a computational cost that grows linearly with image size. A composite loss combining pixel intensity, gradient and structural similarity constraints guides the network to retain thermal saliency, spatial detail and structural consistency simultaneously. Experiments are conducted on three datasets and the proposed method outperforms representative methods, improving average gradient by 38.5% over the second-best method on the infrared-visible satellite dataset and PSNR by 2.1 dB over the second-best method on the near-infrared-visible satellite dataset.

KEYWORDS

Infrared and visible image; Image fusion; Attention mechanism; Transformer; Remote sensing

1. Introduction

Infrared and visible imaging sensors are complementary modalities widely deployed on Earth-observation satellites and unmanned aerial vehicles [1, 32]. Specifically, infrared imagery captures thermal radiation, making heat-emitting targets such as vehicles, ships, and facilities prominently visible across day and night, yet it suffers from low resolution and sparse textural details. Conversely, visible imagery provides rich spatial structures and textures but is severely degraded under low-light or nighttime condi-

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tions [15]. Therefore, fusing these two modalities is essential for robust all-weather remote sensing scene interpretation [41].

Numerous infrared-visible image fusion methods have been proposed, including multi-scale transform [2], sparse representation [3], Convolutional Neural Network (CNN)-based [4], Generative Adversarial Network (GAN)-based [5], and more recently Transformer-based [17, 29, 35] approaches. Despite their progress, they remain constrained by two fundamental limitations. First, traditional methods heavily rely on empirically designed fusion rules, such as activity-level measurements or hand-crafted weight maps [9, 10], which are difficult to generalize across the large radiometric and resolution discrepancies between infrared and visible remote sensing imagery. Second, while recent learning-based methods effectively extract local features, their limited receptive fields lead to inadequate global contextual modeling, often causing the loss of small yet critical objects such as vehicles or vessels in remote sensing scenes [12, 41]. Although Transformer-based architectures address long-range dependencies, they typically process multi-modal features indiscriminately through shared or alternately stacked attention layers [29, 40], introducing substantial cross-modal interference and computational overhead.

To overcome these issues, our key insight is that infrared and visible images contribute fundamentally different types of information to the fusion task. Infrared images primarily provide channel-wise salient intensity (thermal radiation), whereas visible images mostly offer spatially distributed textural details including edges, contours, and structures. Existing attention-based fusion networks do acknowledge that both spatial and channel cues matter, yet they typically compute the two attentions jointly or sequentially on the shared feature stream [12, 45], so the resulting weights are no longer attributable to a specific modality. Feeding these two fundamentally distinct types of information equally into a single network without explicit differentiation inevitably induces modality competition, where one modality may suppress the complementary features of the other. Consequently, we contend that an effective fusion network should: (1) decouple and independently reinforce spatial and channel features according to their respective modality contributions, and (2) globally couple the refined representations to model long-range contextual dependencies, ensuring that small targets are not lost in cluttered backgrounds.

In this paper, we propose an end-to-end fusion network via decoupled parallel attention and global coupling. The network integrates two core components. The Multi-Attention Residual Module (MARM) employs parallel spatial and channel attention branches that separately amplify texture-rich details from the visible image and thermally salient regions from the infrared image, while a residual connection preserves identity information against degradation—this explicitly decouples the modality-specific contributions. The decoupled features are then fed into a Swin Transformer-based module (STM) [25], which utilizes shifted-window self-attention to capture long-range dependencies with linear computational complexity—a process we term global coupling, as it integrates the decoupled local features into a coherent global representation. A composite loss combining intensity, gradient, and structural similarity (SSIM) constraints is further designed to guide the network in retaining complementary information from both modalities. Extensive experiments on three remote sensing datasets demonstrate that the proposed method outperforms representative approaches by a significant margin.

The main contributions of this paper are summarized as follows:

- (1) We propose a decoupled parallel attention mechanism within MARM that explic-

itly resolves modality discrepancies by separately reweighting channel features for infrared thermal saliency and spatial features for visible textures, preventing cross-modal suppression and enhancing complementary information.

- (2) We develop a global coupling strategy that feeds the decoupled, refined features into a STM, enabling efficient long-range dependency modeling with linear computational complexity via shifted-window self-attention.
- (3) We build three benchmark subsets from public remote sensing fusion datasets and demonstrate through extensive experiments that the proposed method outperforms representative approaches on all metrics on the infrared–visible satellite dataset and achieves the best or second-best results on the other two datasets, improving average gradient by 38.5% over the second-best method on the infrared–visible satellite dataset and PSNR by 2.1 dB over the second-best method on the near-infrared–visible satellite dataset.

The remainder of this paper is organized as follows. Section 2 reviews related work on infrared–visible image fusion. Section 3 describes the proposed fusion network. Section 4 gives the experiments conducted to assess the effectiveness of the method. Section 5 concludes the paper.

2. Related Works

Existing infrared and visible image fusion methods have evolved along two dimensions: the way they represent image features, and the way they aggregate cross-modal information.

2.1. *Multi-scale Transform and Sparse Representation in Image Fusion*

Multi-scale transform is one of the most widely used fusion methods based on transform domain, which typically involves three main steps. First, infrared and visible images to be fused are decomposed into different scale representations through multi-scale forward and inverse transforms. Second, fusion rules are designed to select the multi-scale feature coefficients. Finally, fused images are reconstructed via multi-scale inverse transform.

Common multi-scale transforms include Pyramid Transform, Wavelet Transform, Curvelet Transform, and Non-subsampled Transform. The earliest application of the Pyramid Transform in image fusion was the Laplacian Pyramid (LP) [2]. Subsequently, researchers proposed various pyramid-based fusion methods, such as Gradient Pyramid, Contrast Pyramid, Morphological Pyramid, and Steerable Pyramid. Grossmann and Morlet [6] first introduced the concept of Wavelet Transform, and later, Mallat [7] developed the multi-resolution decomposition theory based on Wavelet Transform, which has since been extensively applied in image fusion. Curvelet Transform, derived from both Fourier and Wavelet Transforms, offers superior representation of edge information, making it beneficial for information fusion. Da Cunha et al. [8] proposed the Non-subsampled Contourlet Transform (NSCT), which enables the extraction of salient infrared targets and fine details from visible images, although its computational complexity limits its applicability in real-time scenarios.

Image fusion methods based on sparse representation typically begin by constructing an overcomplete dictionary through learning from a large number of image samples, then applying sparse coefficient decomposition algorithm to represent the source

images sparsely. The fusion process can be summarized as follows. First, image pre-processing decomposes the source images into overlapping blocks or patches for subsequent sparse representation. Second, an overcomplete dictionary is learned from a large collection of high-quality images and sparsely encoding each block or patch to obtain sparse coefficients. Third, the sparse coefficients of infrared and visible images are combined using fusion rules such as maximum selection, window-based methods, or coefficient combination. Finally, the learned dictionary and fused sparse coefficients are used to generate fused images via inverse transform or reconstruction algorithms.

The key of these methods lies in dictionary construction and sparse coding. Dictionary construction is achieved via model-based or learning-based methods. Model-based methods employ fixed transforms such as Discrete Cosine Transform or Wavelet Transform, offering computational simplicity but limited capability for complex data representation. Learning-based methods build overcomplete dictionary from training dataset, providing greater flexibility and effectiveness. Representative dictionary training methods include the Method of Optimal Directions (MOD) and K-Singular Value Decomposition (K-SVD) [9]. MOD seeks to minimize representation error by alternately updating the dictionary and coefficient matrices, however updating the entire dictionary at once obviously slows convergence. K-SVD, derived from Singular Value Decomposition (SVD), iteratively optimizes observed data and updates dictionary matrix so that samples are expressed as combinations of basis vectors, reducing dimensionality while preserving structural information.

In the sparse coding stage, sparse coefficients are computed from the constructed dictionary. Orthogonal Matching Pursuit (OMP), a classic sparse coding method, underpins many compression methods but involves complex fusion rules. The Simultaneous Orthogonal Matching Pursuit (SOMP) algorithm addresses this limitation by improving efficiency while maintaining the same reconstruction error [10], leading to its widespread use in image fusion.

Despite their mathematical elegance, these traditional approaches share a fundamental limitation: the fusion rules are empirically predetermined and remain fixed for all inputs. They cannot adapt to the large radiometric discrepancies between infrared thermal radiation and visible reflectance, nor to the varying resolutions of remote sensing imagery. Consequently, they often introduce artificial artifacts or lose subtle targets in complex battlefield scenes. This rigidity motivated the transition to learning-based paradigms that infer fusion strategies directly from data.

2.2. *Learning-based Methods in Image Fusion*

Traditional image fusion methods often require complex fusion rules, which may impose certain limitations in multimodal image fusion. With the rapid development of computer vision techniques, an increasing number of learning-based image fusion methods have been proposed. Due to their strong capability for feature representation, deep learning-based fusion methods have achieved promising results in infrared–visible image fusion. According to the underlying network architecture, learning-based fusion methods can be broadly categorized into CNN-based methods and GAN-based methods.

Liu et al. [11] were the first to design a CNN-based fusion framework specifically for infrared–visible image fusion. They employed a siamese convolutional network to generate weight maps and directly learned the mapping from multimodal images to fusion weights through the model. Li and Wu [4] proposed a novel image fusion architecture

named DenseFuse, which consists of three core components: encoder, fusion layer, and decoder. The encoder extracts feature maps from infrared and visible images, which are then fused using appropriate fusion strategies, and finally reconstructed into fused images via the decoder. Li et al. [12] introduced NestFuse, a network capable of extracting multi-scale deep features from source images. In this model, a dual-branch attention module was incorporated into the fusion strategy to measure the importance of each spatial position and channel, enabling the fused image to retain critical information from the source images. Xu et al. [13] proposed U2Fusion, an unsupervised end-to-end image fusion network that adaptively preserves source image information by evaluating the importance of extracted features. This approach effectively addresses various fusion tasks, including multi-focus, multi-exposure, and multimodal image fusion. Li et al. [14] developed the Residual Fusion Network (RFN), in which the fusion strategy is implemented via a neural network instead of manually designed rules. Moreover, a novel loss function was introduced to ensure superior fusion performance. Tang et al. [15] proposed the DIVFusion framework, which enhances infrared-visible image fusion under low-light nighttime conditions using an illumination separation network and a texture contrast enhancement network. Zhao et al. [16] introduced the Correlation-Driven Decomposition Fusion Network (CDDFuse), which first employs the Restormer module to capture shallow cross-modal features. A dual-branch structure combining the feature extraction capabilities of Transformer and CNN is then applied to extract global low-frequency information. Meanwhile an Invertible Neural Network (INN) module focuses on extracting local high-frequency details. Finally, the outputs from global and local fusion layers are integrated to generate fused images, thereby improving the performance of downstream tasks such as semantic segmentation and object detection. More recently, Tang et al. [17] developed DATFuse, in which a dual attention Transformer jointly models cross-modal dependencies, further confirming the benefit of attention-based global modelling for this task. Qi et al. [20] proposed DLatLRR-RGF, a saliency-based fusion method that integrates Latent Low-Rank Representation (LatLRR) and Rolling Guidance Filter (RGF) to obtain salient and base components from source images. The salient components are further processed through filtering, and finally fused with the base components to produce fused images.

GAN-based approaches leverage the adaptive learning ability of deep neural networks and have shown remarkable results in various computer vision tasks. Ma et al. [5] proposed FusionGAN, the first GAN-based model for infrared-visible image fusion. In FusionGAN, the generator produces fused images containing the intensity information from infrared images and gradient information from visible images, while the discriminator ensures the preservation of visible image details. This method operates in an end-to-end manner, thus avoiding the design of complex fusion rules. Subsequently, the same group introduced the Dual-Discriminator Conditional GAN (DDcGAN) [18], which addresses the uneven fusion distribution caused by the single adversarial mechanism in FusionGAN. In 2021, they proposed GANMcC [19], a GAN framework with multi-class constraints that transforms image fusion into a multi-distribution estimation problem. By employing multiple classifiers, this method balances the distribution of infrared and visible modalities, thereby enhancing contrast and texture details in fused images.

Overall, learning-based infrared-visible image fusion methods avoid the need for complex fusion rules and demonstrate strong adaptability to various scenarios. However, most CNN-based methods still require source image decomposition, which is often performed using traditional methods, followed by neural networks to guide the

fusion rules. This process results in increased computational complexity. Moreover, while convolution operations excel at extracting local image features, their limited ability to capture global information can lead to partial information loss. GAN-based fusion methods perform well in single-modality fusion tasks but show weaker adaptability in multi-modal tasks. They also lack a balanced consideration of both local and global features of source images, and the convergence is challenging as well.

2.3. *Recent Advances in Attention- and Transformer-based Fusion*

Research on infrared-visible image fusion has accelerated markedly in the past two years, and the recent literature clarifies both the mainstream trends and the questions that remain open. Liu et al. [32] provided a systematic reappraisal of the field, arguing that progress should be measured not only by data compatibility but also by adaption to downstream tasks, and their benchmark confirms that attention-based architectures currently dominate quantitative rankings.

A first line of recent work refines how attention is computed. Zhang et al. [36] proposed FSATFusion, which couples a frequency-domain branch with a spatial attention Transformer so that high-frequency details and global structures are extracted in a single stream. Hu et al. [45] developed AFFusion, in which frequency-integrated spatial and channel attention is combined with atmospheric scattering enhancement to cope with degraded imaging conditions. Yan et al. [40] designed ATFusion, whose alternate cross-attention Transformer lets each modality query the other in successive layers. Liu et al. [33] presented MATCNN, a multi-scale CNN equipped with an attention Transformer that pursues local and global representation jointly. These methods confirm that combining attention with global modelling is the prevailing design choice; however, they either compute spatial and channel attention on a shared feature stream or exchange information between modalities inside every attention layer, so the reweighting is no longer attributable to an individual modality. The decoupled formulation adopted in this paper differs precisely in this respect: spatial and channel responses are refined in parallel and only then coupled globally.

A second line explores alternative global operators. Kang et al. [35] formulated self-attention on the Grassmann manifold to better preserve subspace structure, while state space models have emerged as a competitor to the Transformer. Ma et al. [37] proposed S4Fusion, a saliency-aware selective state space model that captures long-range dependencies with linear complexity, and Zhou et al. [44] combined Mamba and Transformer blocks in MTFusion under semantic segmentation guidance. It should be noted that linear complexity is therefore no longer exclusive to window-based attention; the contribution claimed in this paper is not linear complexity per se, but the explicit ordering of decoupling before global coupling, which is orthogonal to the choice of the global operator.

A third line injects external semantics or task supervision. Guo et al. [39] exploited Segment Anything Model priors in a multi-level collaborative Transformer, Zhang et al. [42] used pretrained vision models to supply semantic priors in SPGFusion, and Yang et al. [43] distilled cross-domain knowledge from high-level vision tasks in KDFuse. Liu et al. [38] coupled fusion with discriminative objectives through cross-dimensional evolutionary learning, and Shi et al. [34] addressed the cross-sensor discrepancy itself via top-K vision alignment. In the remote sensing setting specifically, Li et al. [41] proposed a significance-preserving progressive network to keep small but salient targets from being averaged away during fusion. This line of work is complementary to ours:

it improves what the network is optimized for, whereas the present paper concerns how modality-specific features are organized before global modelling, and the two are combined in the future work outlined in Section 5.

2.4. Motivations and Our Positioning

The above review reveals a clear gap in the literature. On one hand, CNN-based methods excel at local detail extraction but lack global context, leading to small-target loss. On the other hand, Transformer-based methods provide global context but treat all modalities equally, resulting in cross-modal suppression. Crucially, most existing approaches do not systematically decouple channel-wise and spatial-wise complementary information before global modeling and leverages the intrinsic physical distinction between the two modalities: infrared images primarily contribute channel-wise thermal saliency, whereas visible images mainly provide spatially distributed textural details.

Motivated by this observation, we argue that an effective fusion architecture must achieve two objectives that have not been jointly pursued before: (1) Explicit Decoupling—independently reinforce modality-specific attributes (thermal channels vs. spatial textures) to prevent competition; and (2) Global Coupling—model long-range dependencies only after the decoupled, refined features are obtained, so that global attention focuses on meaningful targets rather than irrelevant backgrounds. The proposed network, which integrates a MARM for decoupling and a STM for global coupling, is designed precisely to fill this gap. This explicit separation of decoupling and coupling stages not only preserves complementary information from both modalities but also ensures that global context modeling is performed on clean, modality-distinguished features, achieving superior fusion performance with linear computational complexity.

3. Method

3.1. Architecture Overview

The proposed network, as illustrated in Figure 1, consists of three main stages followed by a reconstruction layer. We consistently denote the spatial height and width of any intermediate 2D feature map (including the input source images) as H and W . Given an infrared image $I_{IR} \in \mathbb{R}^{H \times W}$ and a visible image $I_{VIS} \in \mathbb{R}^{H \times W}$, we first concatenate them along the channel dimension and apply a convolutional layer with a 5×5 convolution, batch normalization and a ReLU activation to extract shallow features $F_{C_1}^{Out}$, which serve as the raw multi-modal input. The shallow features $F_{C_1}^{Out}$ is formulated as:

$$F_{C_1}^{Out} = \text{ReLU} \left(\text{BN} \left(\text{Conv}_5^{2,C} ([I_{IR}, I_{VIS}]) \right) \right) \quad (1)$$

where $\text{Conv}_5^{2,C}(\cdot)$ denotes a 5×5 convolution with 2 input channels and C output channels, and $[\cdot]$ denotes channel-wise concatenation.

Unlike conventional approaches that directly fuse these features or send them indiscriminately into a heavy Transformer, the proposed architecture is specifically driven by the physical characteristics. In remote sensing imagery, infrared targets (e.g., vehicles or ships) are often sparse and occupy only a few pixels, while visible backgrounds contain dense textures. If we apply global self-attention directly on the raw concatenated features, the large number of background pixels dominates the attention com-

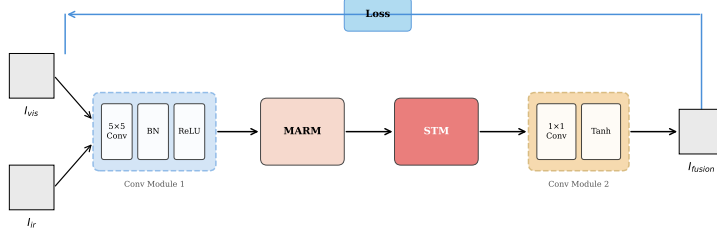


Figure 1. Image fusion network architecture based on Multi-attention mechanism and Transformer.

putation, drowning out small thermal targets. Therefore, we place the MARM at the front end as a decoupling stage.

After decoupling by MARM, redundant background responses are largely suppressed, and the feature map retains only the most informative, modality-specific cues. Feeding these decoupled features into the subsequent STM allows the shifted-window self-attention to focus on establishing long-range dependencies among meaningful targets—for example, linking a distant vehicle to the road lane, or connecting a ship to its wake—rather than wasting computation on irrelevant background patches. This stage is termed global coupling, as it integrates the locally refined features into a coherent global representation that preserves spatial coherence across the entire scene. The Swin Transformer architecture ensures that this global modeling is achieved with linear computational complexity, making the approach scalable to high-resolution remote sensing imagery. The final reconstruction convolution then compresses the channel dimension to produce the fused image I_{Fused} .

3.2. Multi-attention Residual Module

Human vision concentrates on a limited region of the visual field at any given moment, which reduces interference and prioritizes important information. The multi-attention mechanism is introduced so that the fusion network mimics this behavior, focusing more precisely on the key positions and salient content of an image while keeping the model compact. Infrared remote sensing images carry pronounced thermal radiation information, whereas visible remote sensing images carry prominent background texture; the two modalities differ markedly in appearance, yet remain complementary. Most existing deep learning-based fusion methods treat spatial and channel features of both modalities equally and cannot bias computational resources towards important regions. Simply deepening the network to strengthen feature representation increases computational and storage cost.

To address these problems, we design the MARM, which comprises a residual block together with coordinate, channel and spatial attention blocks, so that the extraction and expression of important features is enhanced without the overhead of a deeper network. The module architecture is shown in Figure 2. After low-level features are extracted from the infrared and visible images, they are fed into MARM to represent the important local information of the scene.

The Coordinate Attention Block (CA) feature extraction process is expressed as:

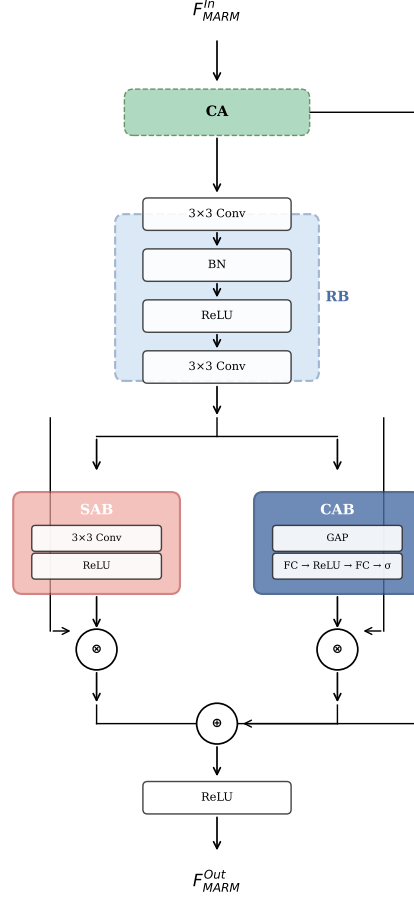


Figure 2. Multi-attention residual module architecture.

$$F_{CA}^{Out} = CA(F_{C_1}^{Out}) \quad (2)$$

The Residual Block (RB) feature extraction process is defined as:

$$F_{RB}^{Out} = RB(F_{CA}^{Out}) \quad (3)$$

The Spatial Attention Block (SAB) feature extraction process is defined as:

$$F_{SAB}^{Out} = SAB(F_{RB}^{Out}) \quad (4)$$

The Channel Attention Block (CAB) feature extraction process is expressed as:

$$F_{CAB}^{Out} = CAB (F_{RB}^{Out}) \quad (5)$$

The final output of MARM is formulated as:

$$F_{MARM}^{Out} = \text{ReLU} (F_{CAB}^{Out} \cdot F_{RB}^{Out} + F_{SAB}^{Out} \cdot F_{RB}^{Out} + F_{CA}^{Out}) \quad (6)$$

where $\cdot$ denotes element-wise multiplication. The output of the residual block is reweighted by the channel and spatial attention maps respectively, the two reweighted terms are summed together with the coordinate attention output as a residual short-cut, and a ReLU activation is applied to the sum.

3.2.1. Coordinate Attention Block

The coordinate attention block [21] effectively encodes channel relationship and long-range dependencies of images through precise position information, which mainly includes coordinate information embedding and coordinate attention generation.

For coordinate information embedding, given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, the block decomposes global pooling into two separate 1D encodings. Specifically, pooling kernels with size of $(H, 1)$ and $(1, W)$ are applied to encode each channel along the horizontal and vertical direction, respectively. The resulting aggregated tensors are formulated as:

$$\begin{cases} z_c^h(h) = \frac{1}{W} \sum_{0 \leq i < W} x_c(h, i), \\ z_c^w(w) = \frac{1}{H} \sum_{0 \leq j < H} x_c(j, w). \end{cases} \quad (7)$$

where $x_c(h, i)$ denotes the pixel value of the c -th channel at spatial location (h, i) with height index h and width index i ; similarly, $x_c(j, w)$ denotes the value at location (j, w) . The terms $z_c^h(h)$ and $z_c^w(w)$ denote the output receptive attention maps in the horizontal and vertical directions, respectively.

For coordinate attention generation, attention maps generated in the previous step are firstly concatenated and transformed using a shared convolution with a kernel size of 1×1 , as denoted below:

$$f = \delta \left(\text{Conv}_{1 \times 1} \left(\left[Z^h, Z^w \right] \right) \right), \quad (8)$$

where f denotes the intermediate feature map of spatial information in both horizontal and vertical directions, $\delta(\cdot)$ denotes the non-linear activation function, $\text{Conv}_{1 \times 1}(\cdot)$ denotes the 1×1 convolution operation, $[\cdot]$ denotes the concatenation operation, Z^h and Z^w denote the aggregated attention maps in the horizontal and vertical directions, respectively.

Secondly, f is divided into two independent feature maps f^h and f^w along the spatial dimension. Then two 1×1 convolutions are applied to adjust the channels of

the feature maps f^h and f^w to match with input X . The detailed process is formulated as follows:

$$\begin{cases} g^h = \sigma(\text{Conv}^{1 \times 1}(f^h)) \\ g^w = \sigma(\text{Conv}^{1 \times 1}(f^w)) \end{cases} \quad (9)$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, while g^h and g^w denote the attention weights in the horizontal and vertical directions, respectively.

Finally, the attention weights of the c -th channel are multiplied with the corresponding coordinate position to compute the attention map in the horizontal and vertical directions. The final output of the coordinate attention block is obtained, which is denoted as:

$$y_c(i, j) = x_c(i, j) g_c^h(i) g_c^w(j), \quad (10)$$

where $y_c(i, j)$ denotes the final feature map computed by the coordinate attention block.

3.2.2. Spatial Attention Block

The spatial attention mechanism adaptively selects and optimizes spatial regions by capturing the spatial relationships between feature points, thereby strengthening feature representation for remote sensing imagery. In infrared-visible image fusion, the quality of the fused image depends largely on the regions that are directly relevant to the task: typically the thermal radiation information in the infrared image and the background detail in the visible image, especially at salient targets or specific regions. The core objective of spatial attention is to assign appropriate weights to these regions of interest and feed the weighted result back into the feature map. This is particularly critical in fusion tasks, as it produces feature maps in which essential information is markedly enhanced and less relevant information is suppressed.

The proposed SAB consists of convolutional modules, ReLU activation function, and Sigmoid activation function. It is designed to extract spatial features from multimodal remote sensing images. The feature extraction process is formulated as follows:

$$F_{SAB}^{Out} = \sigma \left(\text{Conv}_3^{C/R, 1} \left(\text{ReLU} \left(\text{Conv}_3^{C, C/R} (F^{In}) \right) \right) \right) \quad (11)$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, $\text{Conv}_3^{C/R, 1}(\cdot)$ denotes a 3×3 convolution with input channels C/R and output channel 1, $\text{ReLU}(\cdot)$ denotes the ReLU activation function, $\text{Conv}_3^{C, C/R}(\cdot)$ denotes a 3×3 convolution with input channels C and output channels C/R , F^{In} denotes the input feature map, and F_{SAB}^{Out} denotes the output feature map extracted by SAB.

3.2.3. Channel Attention Block

In deep learning, the channel attention mechanism adaptively assigns weights to each feature channel based on its importance, enabling the model to focus more on informa-

tive channels. This is achieved by automatically learning the relevance of each channel and adjusting their contributions accordingly. For the infrared–visible image fusion task, the channel attention mechanism aims to enhance the feature channels that are beneficial to the fusion task while suppressing those that contribute less. The proposed CAB is based on the Squeeze-and-Excitation (SE) Block [22], which explicitly models inter-channel dependencies and adaptively recalibrates channel-wise feature responses. During the squeeze operation, the spatial dimension ($H \times W$) of the input feature map $U \in \mathbb{R}^{C \times H \times W}$ is aggregated to generate channel-wise descriptors. This is achieved via global average pooling, which compresses each 2D feature map into a single scalar, thereby transforming the tensor map from $C \times H \times W$ to $C \times 1 \times 1$. The detailed process is formulated as follows:

$$F_{sq}(u_c) = \frac{1}{H \times W} \sum_{i=1}^W \sum_{j=1}^H u_c(i, j) \quad (12)$$

where $u_c(i, j)$ denotes the value of the corresponding pixel in the feature map.

The excitation operation fully leverages the aggregated information obtained from the squeeze step. It employs a simple gating mechanism with Sigmoid activation function to capture channel-wise dependencies. The detailed process is formulated as follows:

$$s = \sigma(W_{fc2} \text{ReLU}(W_{fc1}z)) \quad (13)$$

where s denotes the learned channel attention, $\sigma(\cdot)$ denotes the Sigmoid activation function, $\text{ReLU}(\cdot)$ denotes the ReLU activation function, while W_{fc1} and W_{fc2} denote the fully connected layers. After normalization by Sigmoid activation function, the output encodes the channel-wise attention. This attention is then applied to the original feature map through element-wise multiplication, adaptively enhancing or suppressing feature responses in each channel based on task relevance.

3.2.4. Residual Block

To fully capture the complementary information that might be lost during convolution operations, the residual block is constructed. This block aims to maximize the preservation of both prominent thermal information from infrared images and fine-grained texture details from visible images in the fusion task. The residual block consists of two convolutional modules, and the specific process is formulated as follows:

$$F_{RB}^{Out} = BN \left(Conv_3^{C,C} \left(\text{ReLU} \left(BN \left(Conv_3^{C,C} (F^{In}) \right) \right) \right) \right) \quad (14)$$

where $BN(\cdot)$ denotes the batch normalization, $Conv_3^{C,C}(\cdot)$ denotes a 3×3 convolution with input and output channels C , $\text{ReLU}(\cdot)$ denotes the ReLU activation function, and F^{In} denotes the input feature map. To prevent losing important information, residual connections are constructed between the residual block and the parallel spatial and channel attention blocks.

3.3. Swin Transformer-based Module

In multimodal remote sensing image fusion, capturing global context is of paramount importance. Relying solely on CNNs to model global semantic context remains challenging, because the local nature of the convolution operation limits the ability to capture long-range dependencies. The success of Transformer models [23] in natural language processing has introduced a new paradigm for modeling global relationship in vision tasks, and the Vision Transformer [24] demonstrated that pure attention architectures are competitive for image recognition. In 2021, Microsoft Research proposed the Swin Transformer [25], which achieved remarkable performance across various computer vision benchmarks. Researchers have also successfully applied the Swin Transformer in medical image processing [26]. However, studies exploring the direct application of Transformer-based models to multimodal remote sensing image fusion remain limited. The Swin Transformer offers four model variants. In this paper, the proposed Transformer module is built upon the Swin-T variant and specifically leverages two blocks from Stage 1. By employing window-based multi-head self-attention (W-MSA) and shifted window-based multi-head self-attention (SW-MSA), the module captures global information while maintaining linear computational complexity. It adopts a pure self-attention mechanism to extract high-level semantic information, which enables global context to be captured and detailed information from both modalities to be preserved, and consequently improves the quality of the fused image.

The architecture of the proposed module is illustrated in Figure 3. The feature maps obtained from the MARM are fed into the Transformer module. First, the feature maps $F_{MARM}^{Out} \in \mathbb{R}^{C \times H \times W}$ are fed into a patch partition module. This module splits the spatial grid into non-overlapping patches, reshaping the tensor into $N \times (P^2 \times C)$, where P is the patch size and $N = \frac{H}{P} \times \frac{W}{P}$ is the number of patches. Next, a linear embedding layer projects each patch to an embedding dimension C , yielding a feature map of shape $C \times \frac{H}{P} \times \frac{W}{P}$. These embedded features are then passed through two consecutive Transformer blocks with enhanced self-attention mechanism. In the first block, the features undergo Layer Normalization (LN) layer before being fed into the W-MSA module. The output of W-MSA is added back to the input through a residual connection. The result is again normalized by LN and passed through a Multi-layer Perceptron (MLP), followed by another residual connection to complete the first block. The second block shares a similar structure, except that W-MSA is replaced with the SW-MSA module to enhance cross-window communication. The output of this block serves as the final representation extracted by the Transformer module. The detailed process is formulated as follows:

$$\begin{aligned}
F_{SA}^1 &= \text{W-MSA} \left(\text{LN} \left(F_{MARM}^{Out} \right) \right) + F_{MARM}^{Out} \\
F_{STM}^{Out1} &= \text{MLP} \left(\text{LN} \left(F_{SA}^1 \right) \right) + F_{SA}^1 \\
F_{SA}^2 &= \text{SW-MSA} \left(\text{LN} \left(F_{STM}^{Out1} \right) \right) + F_{STM}^{Out1} \\
F_{STM}^{Out2} &= \text{MLP} \left(\text{LN} \left(F_{SA}^2 \right) \right) + F_{SA}^2
\end{aligned} \tag{15}$$

where F_{SA}^1 and F_{SA}^2 denote the result of residual connections of two Blocks, respectively. W-MSA($\cdot$) denotes the W-MSA computation within windows, F_{MARM}^{Out} denotes the input feature map to the Transformer module, F_{STM}^{Out1} denotes the output of the first Transformer block, and SW-MSA($\cdot$) denotes self-attention computation for neighboring windows using SW-MSA.

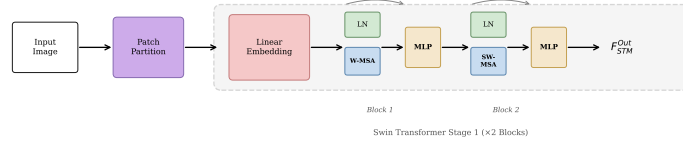


Figure 3. Swin Transformer-based module architecture.

3.3.1. *W-MSA and SW-MSA*

The Transformer module first partitions the input feature map into a series of uniformly sized, non-overlapping windows. Self-attention is then computed independently within each window, which significantly reduces the computational cost. Given an input feature map of shape $C \times H \times W$, the computational complexity of the self-attention mechanism in this module is computed as:

$$\Omega(\text{MSA}) = 4HWC^2 + 2(HW)^2C \quad (16)$$

Assuming each non-overlapping window has a height and width of M (with $M = 7$ in our Swin-T configuration), the entire feature map is divided into $\lceil H/M \rceil \times \lceil W/M \rceil$ windows, where the cyclic-shifting mechanism in SW-MSA handles the boundary cases when H or W is not a multiple of M . The computational complexity of self-attention within a single window is computed as:

$$\Omega(W) = 4M^2C^2 + 2M^4C \quad (17)$$

Considering the total number of windows, the overall computational complexity of W-MSA is computed as:

$$\Omega(\text{W-MSA}) = 4HWC^2 + 2M^2HWC \quad (18)$$

where Ω denotes the computational complexity. It is evident that the complexity of W-MSA grows linearly with the spatial size of the input feature map. In contrast, the traditional MSA mechanism exhibits quadratic complexity with respect to the spatial dimensions. Therefore, W-MSA significantly reduces the computational complexity and provides a clear advantage in terms of efficiency.

While W-MSA significantly reduces computational complexity by computing self-attention within non-overlapping windows, it inherently limits global modeling capability due to the lack of cross-window information interaction. To address this limitation, the SW-MSA mechanism is introduced, which enhances inter-window communication while preserving the computational efficiency of non-overlapping windows.

As illustrated in Figure 4, Figure 4(a) represents the W-MSA module, and Figure 4(b) shows the SW-MSA module. Unlike W-MSA, SW-MSA introduces an offset to the window partitioning, enabling adjacent windows in W-MSA to share information. For instance, the second window in the first row of Figure 4(b) allows interaction between the two adjacent windows in the first row of Figure 4(a). Similarly, the first

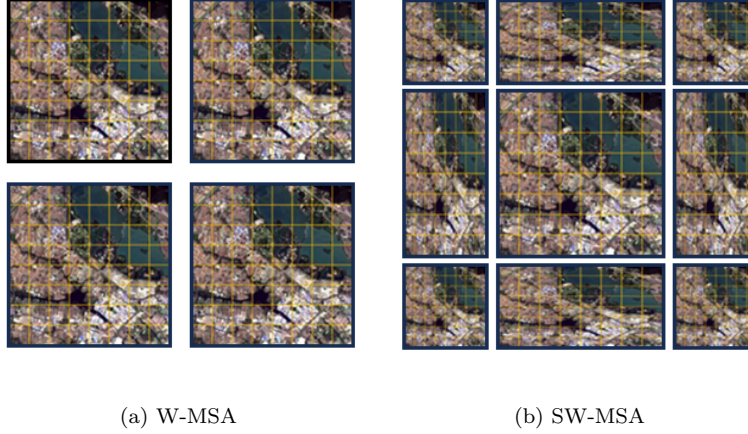


Figure 4. W-MSA and SW-MSA.

window in the second row of Figure 4(b) allows interaction between the two adjacent windows in the first column of Figure 4(a).

The shifting strategy of SW-MSA increases the total number of windows and results in some windows having smaller sizes than $M \times M$, slightly increasing the computation cost. To address this issue, the cyclic shifting method is applied. Specifically, the feature map is shifted diagonally towards the top-left corner, regrouping spatially distant sub-windows into a batch of new windows. This cyclic shifting method ensures that the number of windows remains the same as the original partitioning (e.g., four windows before and after the cyclic shifting) while enabling interaction across window boundaries.

To prevent mutual interference during self-attention computation across different sub-windows, the masked MSA mechanism is adopted, ensuring that self-attention is strictly confined within each sub-window. This design enables effective global representation learning while maintaining linear computational complexity.

3.3.2. Masked MSA Mechanism

The computation of self-attention within both W-MSA and SW-MSA modules for each attention head is formulated as:

$$\text{Attention}(Q, K, V) = \text{SoftMax} \left(\frac{QK^T}{\sqrt{d}} + B \right) V \quad (19)$$

where Attention denotes the computed weights, Q, K, V denote the query, key, and value matrices, respectively. B represents a learnable relative position bias matrix that encodes the positional relationships between tokens, and d denotes the sequence feature dimension.

As illustrated in Figure 5, Figure 5(a) shows four non-overlapping windows used in the standard W-MSA computation, where self-attention is independently computed within each window. Figure 5(b) presents the windows after a right-down shift, enabling the SW-MSA mechanism to capture cross-window interactions. To better illustrate window correspondence, Figure 5(c) uses numerical labels: for example, window

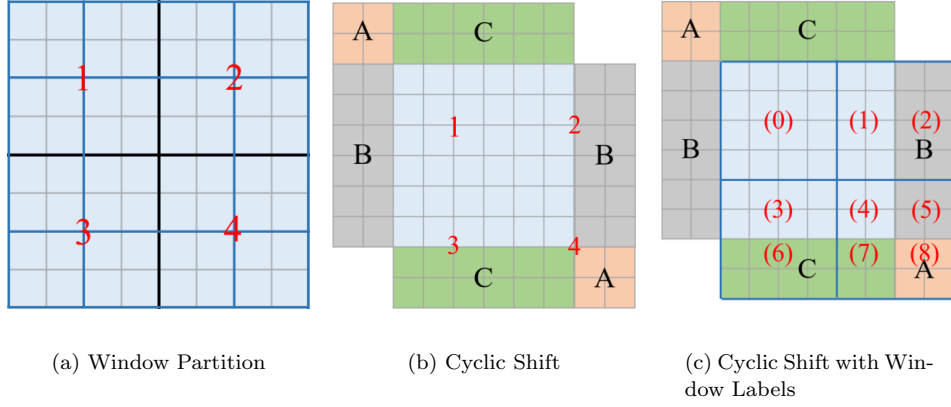


Figure 5. Sliding window mechanism.

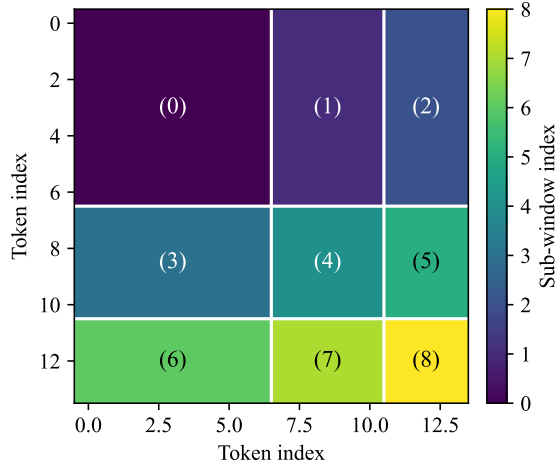


Figure 6. Masked MSA mechanism.

(0) in Figure 5(c) directly corresponds to window (1) in Figure 5(b), and thus allows direct self-attention computation. However, windows such as (1) and (2) in Figure 5(c), which are not adjacent in the original feature map, could not compute cross-window self-attention jointly. Similarly, window pairs like (3)(6) or (4)(5)(7)(8) should not be involved in cross-window computation. Based on this, a masked MSA mechanism is applied. This mechanism operates by subtracting and assigning values. Pixels within the same window receive a zero mask. However, invalid attention positions are assigned a large negative value (e.g., -100) before softmax so that their attention weights become zero. The masking result is visually demonstrated in Figure 6.

3.4. Loss Function

In order to ensure that the fusion result both highlights infrared thermal radiation information and retains rich texture detail, a composite loss function consisting of pixel intensity loss, gradient loss and structural loss is designed and combined in a weighted form. The pixel intensity loss constrains the pixel intensity distribution of the fused image with respect to the source images, that is, the texture information of

the visible image and the radiation information of the infrared image. The gradient loss constrains spatial detail and captures salient detail through maximum aggregation of gradient information. The structural loss constrains the structural similarity between the fusion result and the source images, and evaluates whether the result is distorted in terms of luminance, contrast and structure. Weighting factors control the contribution of each term. The pixel intensity loss is defined as:

$$\mathcal{L}_P = \frac{1}{H \times W} \|I_F - I_{IR}\|_F^2 + \alpha \cdot \frac{1}{H \times W} \|I_F - I_{VIS}\|_F^2 \quad (20)$$

where $I_F \in \mathbb{R}^{H \times W}$ denotes the fused remote sensing image, and $\|\cdot\|_F$ denotes the Frobenius norm, and α is the weight.

The gradient loss is defined as:

$$\mathcal{L}_G = \|\nabla I_F - \max\{\nabla I_{IR}, \nabla I_{VIS}\}\|_2 \quad (21)$$

where ∇ denotes the gradient operation, $\max\{\cdot, \cdot\}$ denotes taking the maximum value, and $\|\cdot\|_2$ denotes the L2 norm.

The structural loss is defined as:

$$\mathcal{L}_S = 1 - \text{SSIM}(I_F, \max\{I_{IR}, I_{VIS}\}) \quad (22)$$

where $\text{SSIM}(\cdot)$ is the structural similarity measure.

Finally, the overall loss is defined as:

$$\mathcal{L} = \mathcal{L}_P + \lambda \cdot \mathcal{L}_G + \gamma \cdot \mathcal{L}_S \quad (23)$$

where λ and γ are the weights controlling the gradient loss and structural loss, respectively.

4. Experiments

To evaluate the effectiveness of the proposed method, fusion experiments are conducted on infrared and visible satellite remote sensing datasets. Five representative methods are selected for comparison, including LatLRR [27], AUIF [28], DenseFuse [4], FusionGAN [5], and SwinFusion [29]. For the infrared and visible aerial dataset and the near-infrared and visible (NIR-VIS) satellite dataset, three methods (AUIF, FusionGAN, and SwinFusion) are used for comparison. LatLRR and DenseFuse are excluded there because their released implementations could not be fairly adapted to the remote sensing data scales and the training pipeline adopted in this paper.

4.1. Datasets

Since existing public datasets cannot directly meet the requirements of infrared-visible remote sensing image fusion research, we compile three paired subsets from public

satellite and aerial sources. The three datasets are summarized as follows.

- (1) **Infrared–visible satellite dataset (IV-Sat).** The source images are derived from Landsat 8, which was launched by NASA in November 2013 and carries the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). Bands 2, 3 and 4 are composited into true-color visible images with a spatial resolution of 30 m, while band 10 is taken as the infrared image. Band 10 has a native resolution of 100 m and is therefore resampled to 30 m and co-registered with the visible bands before patch extraction.
- (2) **Infrared–visible aerial dataset (IV-Air).** The images are selected from the DroneVehicle dataset [31] released by Tianjin University after scene screening. This dataset provides drone-borne RGB–infrared image pairs of ground vehicles acquired under both daytime and night-time conditions, and is therefore representative of aerial reconnaissance scenarios.
- (3) **Near-infrared–visible satellite dataset (NIR-VIS-Sat).** The images are selected from the WHU-OPT-SAR dataset [30] released by Wuhan University, which contains 100 optical images acquired by the GF-1 satellite and covers diverse terrains such as plains, forests and mountains. Only the visible and near-infrared bands are used in this paper.

For all three datasets, the visible (or true-color composited) images are converted to single-channel grayscale images before being fed into the network. To enlarge the training set and to simulate the mild residual misregistration that is unavoidable in real multi-sensor acquisitions, each image pair is synthesized as follows. A square patch is randomly cropped from the visible image, and its four corners are randomly perturbed within $[-25, 25]$ pixels. The homography matrix computed from the four corner correspondences is inversely applied to the infrared image, and a patch of the same size and position is then cropped from the transformed result. The perturbation magnitude is small relative to the patch size, so the synthesized pairs remain approximately aligned while providing geometric diversity. Following this procedure, 4000 image pairs are generated for each of the three datasets and divided into training and test sets at a ratio of 9:1, yielding 3600 training pairs and 400 test pairs.

4.2. *Experimental Settings*

The fusion experiments are conducted using PyTorch framework on a workstation running Ubuntu 20.04 operating system equipped with an Intel(R) Xeon(R) Silver 4216 CPU and an NVIDIA GeForce RTX 3080 GPU. The network is trained with the Adam optimizer. The model is trained for 20 epochs with a batch size of 8 and an initial learning rate of 0.0001. The weighting factors of the composite loss function are set to $\alpha = 0.5$, $\lambda = 0.5$ and $\gamma = 0.5$. For comparison, AUIF, DenseFuse, FusionGAN, and SwinFusion are implemented using PyTorch framework, while LatLRR is implemented in MATLAB R2016a on a workstation running Windows 11 operating system equipped with an Intel Core i5-12500H 2.50GHz CPU.

4.3. *Comparison with Representative Methods*

4.3.1. *Quantitative Evaluation and Analysis*

Five widely used metrics are adopted for quantitative evaluation: spatial frequency (SF), average gradient (AG), PSNR, mean squared error (MSE), and visual informa-

tion fidelity (VIF). The values in Tables 1–3 represent the average scores across all test samples for each dataset. For clarity, $\uparrow$ means that higher values indicate better fusion performance, and $\downarrow$ means that lower values indicate better fusion performance. The best results are underlined, and the second-best results are marked in bold.

On the IV satellite dataset (Table 1), the proposed method achieves the best result on all five metrics. It attains an SF of 35.367, which is 9.8% higher than the second-best value of SwinFusion (32.222), and an AG of 28.462, which is 38.5% higher than the second-best value of LatLRR (20.553). For the fidelity-related metrics, the proposed method obtains a PSNR of 64.897 dB, an MSE of 0.025, and a VIF of 0.622, all of which are better than the corresponding second-best values (64.233 dB, 0.029, and 0.610, respectively). It is also worth noting that FusionGAN performs markedly worse than the other methods on this dataset (SF of 4.949 and VIF of 0.114), suggesting that the single-generator adversarial training is not well suited to satellite remote sensing scenes.

On the IV aerial dataset (Table 2), the proposed method ranks first on four of the five metrics and second on VIF. The largest relative gain is observed for AG, where the proposed method reaches 9.463, which is 13.5% higher than the second-best value of SwinFusion (8.337). It also obtains the best SF of 22.907 (vs. 22.754, +0.7%), the best PSNR of 63.715 dB (vs. 62.765 dB, +0.95 dB), and the lowest MSE of 0.037 (vs. 0.039). On VIF, the proposed method reaches 0.586, which is slightly lower than SwinFusion (0.600) but still clearly above AUIF (0.420) and FusionGAN (0.176).

On the NIR-VIS satellite dataset (Table 3), the proposed method achieves the best results on four metrics and ranks second on SF. The AG reaches 8.439, exceeding the second-best value of AUIF (7.269) by 16.1%. The PSNR of 69.862 dB is 2.1 dB higher than SwinFusion (67.738 dB), and the MSE of 0.009 is lower than the second-best MSE of 0.014. The VIF of 0.950 slightly exceeds AUIF (0.948). For SF, the proposed method obtains 18.973, which is second only to AUIF (25.310); this indicates that AUIF preserves a higher spatial-frequency content on this dataset, but the proposed method still provides the best overall trade-off according to the other four metrics.

Taken together, the quantitative results on the three datasets show that the proposed method achieves the best performance on all five metrics on the IV satellite dataset, and the best or second-best performance on every metric on the other two datasets, with particularly large gains in AG (up to 38.5%) and PSNR (up to 2.1 dB). The consistent improvement over the CNN-based (DenseFuse, AUIF), GAN-based (FusionGAN), and Transformer-based (SwinFusion) baselines indicates that the decoupled parallel attention and global coupling mechanism generalizes well across different sensors and imaging conditions.

Table 1. Experimental results of infrared and visible satellite remote sensing images

Fusion Method	SF $\uparrow$	AG $\uparrow$	PSNR $\uparrow$	MSE $\downarrow$	VIF $\uparrow$
LatLRR	18.492	20.553	59.424	0.078	0.610
AUIF	31.657	13.054	63.264	0.035	0.609
DenseFuse	21.152	8.870	64.233	0.029	0.529
FusionGAN	4.949	1.937	61.945	0.047	0.114
SwinFusion	32.222	13.126	61.456	0.051	0.581
Proposed	<u>35.367</u>	<u>28.462</u>	<u>64.897</u>	<u>0.025</u>	<u>0.622</u>

Table 2. Experimental results of infrared and visible aerial remote sensing images

Fusion Method	SF↑	AG↑	PSNR↑	MSE↓	VIF↑
AUIF	16.772	5.547	58.182	0.101	0.420
FusionGAN	2.999	0.954	62.765	0.039	0.176
SwinFusion	22.754	8.337	62.519	0.042	<u>0.600</u>
Proposed	<u>22.907</u>	<u>9.463</u>	<u>63.715</u>	<u>0.037</u>	0.586

Table 3. Experimental results of near-infrared and visible satellite remote sensing images

Fusion Method	SF↑	AG↑	PSNR↑	MSE↓	VIF↑
AUIF	<u>25.310</u>	7.269	60.738	0.056	0.948
FusionGAN	2.666	1.009	66.125	0.019	0.198
SwinFusion	17.241	7.116	67.738	0.014	0.932
Proposed	18.973	<u>8.439</u>	<u>69.862</u>	<u>0.009</u>	<u>0.950</u>

4.3.2. Qualitative Evaluation and Analysis

As illustrated in Figure 7, the qualitative fusion results of six methods on infrared and visible satellite remote sensing images are presented. Compared to other fusion methods, the proposed method effectively integrates complementary information from both infrared and visible remote sensing images across five representative remote sensing scenarios. The fused images not only preserve the rich spatial details of visible images but also incorporate the salient thermal radiation features from infrared images.

In the first and third columns, representing farmland and village scenes respectively, the proposed method yields clearer boundaries of thermally prominent targets. In the second column (harbor scene), the fused image presents more natural details while enhancing the display of roads originally visible better in visible images. For the fourth column (river scene), the proposed method preserves the background clarity of visible images and highlights targets with emphasized thermal radiation information such as ships. In the fifth scene, where the water boundaries are indistinct in the visible images, the fused image effectively retains the superior thermal radiation information from the infrared image. Compared with other fusion methods, the proposed method generates results that maintain a high degree of visual fidelity and consistency with the source images, demonstrating its superior performance in preserving both spatial structure and semantic content.

Figure 8 presents the fusion results of four methods applied to infrared and visible aerial remote sensing images. Compared to other methods, the proposed method demonstrates superior detail preservation while effectively incorporating salient thermal radiation information from the infrared modality. This enhancement is particularly beneficial for downstream tasks such as object detection and image segmentation.

Due to the inherent differences in imaging mechanisms and resolutions between visible and infrared sensors, visible images in the first three columns clearly capture features such as road divider markings (the first column), numerical symbols on the ground (the second column), and rooftop structures of vehicles (the third column), which are not visible in the corresponding infrared images. In contrast, the images in

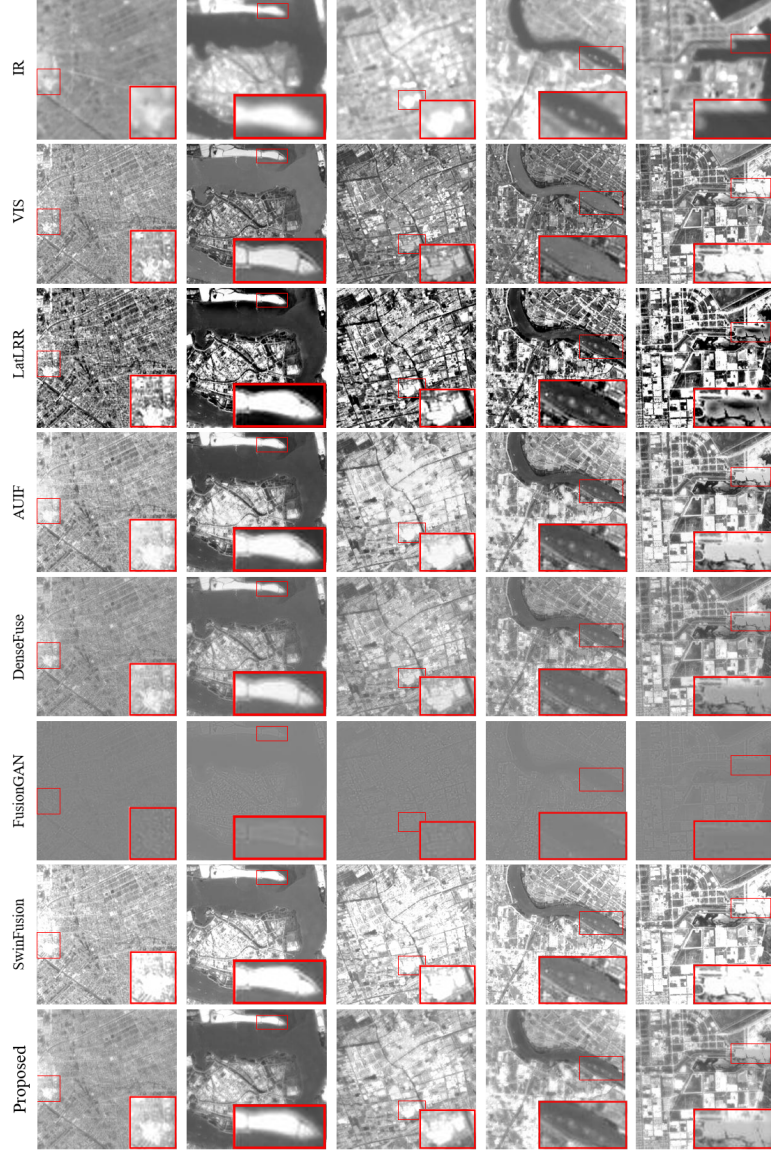


Figure 7. Infrared and visible satellite remote sensing image fusion results.

the fourth and fifth columns highlight the advantages of infrared sensing: thanks to thermal radiation properties, targets such as ground vehicles remain distinguishable even under nighttime low-light conditions, whereas the visible sensor fails to capture meaningful information.

Although all four methods fuse complementary information to some extent, the proposed method consistently achieves better preservation of fine details—such as road markings, ground numerical symbols, and vehicle structural information—while maintaining fidelity to the source images. In nighttime low-light conditions, the proposed method not only enhances vehicle targets’ visibility but also retains richer background context.

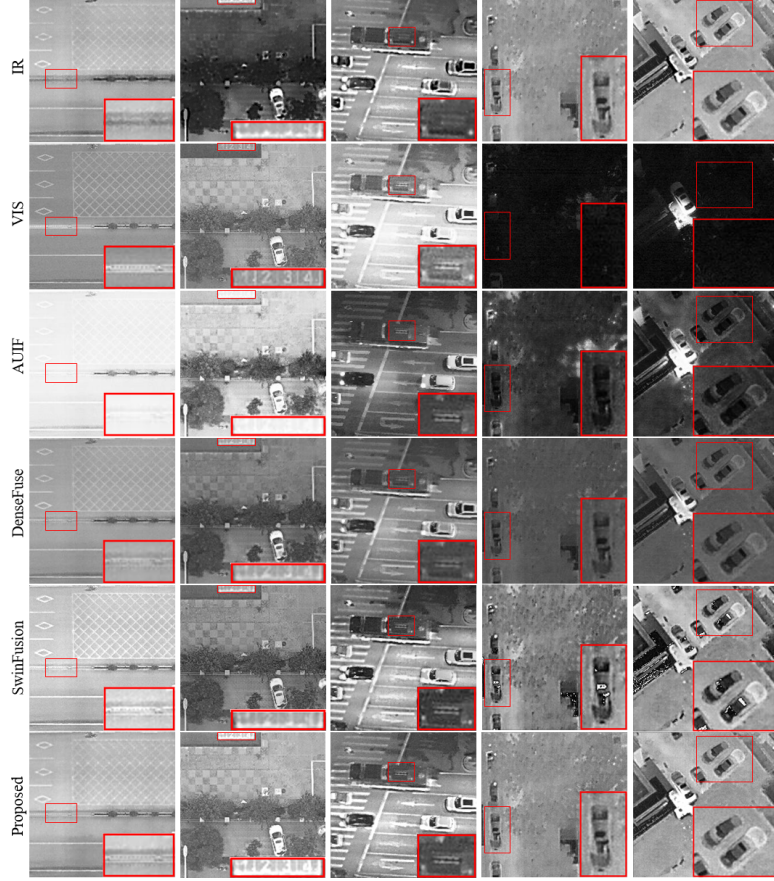


Figure 8. Infrared and visible aerial remote sensing image fusion results.

4.4. Ablation Studies

4.4.1. Ablation of the Modules

As illustrated in Figure 9, removing the STM leads to a noticeable degradation in performance, particularly because the model loses its ability to capture long-range dependencies. This manifests as blurred regions in the fused image of Figure 9(c). Conversely, removing the MARM reduces the capability to extract fine-grained details and to suppress irrelevant information, making it difficult for the model to concentrate on salient features, as shown in Figure 9(d). Compared with both ablated variants, the complete model in Figure 9(e) preserves the thermal saliency of the infrared input in Figure 9(b) together with the texture of the visible input in Figure 9(a). These findings indicate that the proposed method, which integrates both MARM and STM, captures local salient structures and global context simultaneously, and that the absence of either module degrades performance, highlighting their complementary roles.

To further quantify the contribution of each module, we conduct a comparative evaluation between the full model and its two ablated variants on the constructed dataset. The quantitative results, presented in Tables 4, 5 and 6, are assessed using five widely-used image fusion metrics. The best results are underlined for clarity. Among all metrics, the full model consistently outperforms the ablated versions, confirming the effectiveness of the MARM and STM modules in enhancing overall fusion quality. It is noteworthy that removing either module causes a dramatic drop in AG and SF.

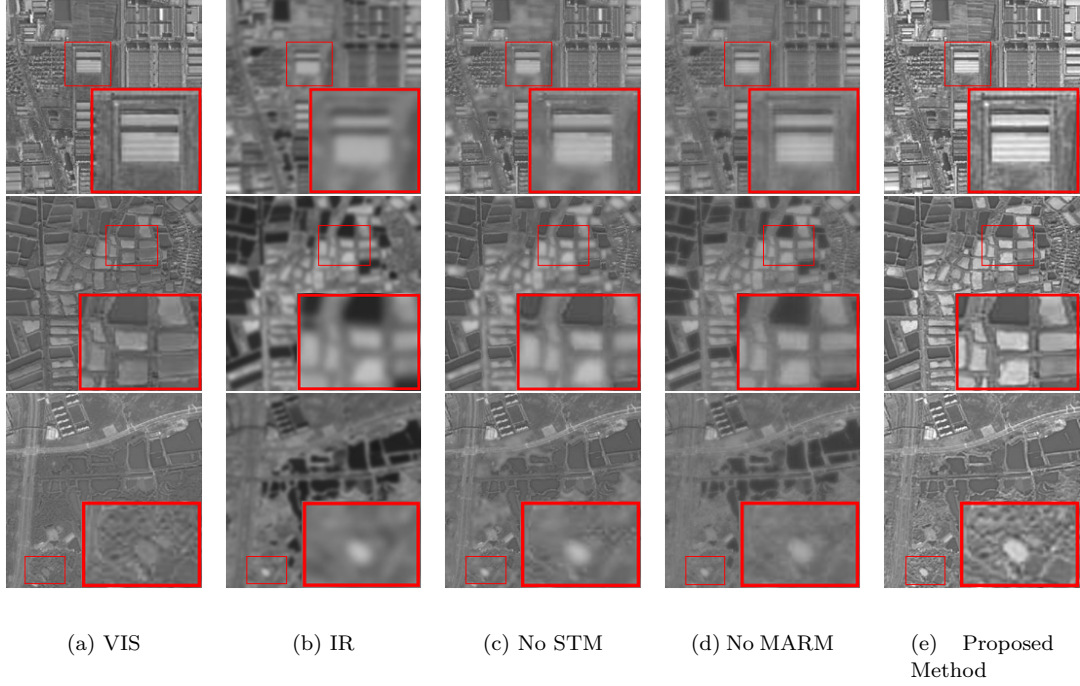


Figure 9. Module ablation experiment fusion results.

This indicates that MARM and STM are not merely additive components but are strongly coupled in our architecture: MARM conditions the features for STM, and STM in turn globally integrates them. The absence of either breaks this cooperative mechanism, leading to significant performance degradation.

4.4.2. Ablation of the Loss Function

To further validate the necessity of each loss within the proposed loss function which comprises pixel intensity loss, gradient loss, and structural loss, we conduct an ablation study by designing six configurations. The six configurations are as follows: pixel intensity loss only, gradient loss only, structural loss only, pixel intensity and gradient loss, pixel intensity and structural loss, gradient and structural loss. Each configuration removes one or more losses from the loss function and retrain the fusion network on infrared and visible aerial remote sensing images. Representative fusion results are presented in Figure 10, in which Figures 10(a) and 10(b) give the visible and infrared source images for reference, and regions of interest are highlighted using red rectangles.

The experimental results reveal that excluding pixel intensity loss leads to a noticeable decline in the network’s ability to capture prominent features: in Figures 10(d), 10(e) and 10(h), where this term is absent, the vehicle targets within the highlighted regions appear significantly blurred relative to the result obtained with the complete loss in Figure 10(i). When gradient loss is removed, the fused images tend to lose important texture details and exhibit visible artifacts, suggesting a diminished ability to preserve edge and structure information, as Figures 10(c), 10(e) and 10(g) show. Furthermore, the absence of structural loss results in distortions in image brightness, contrast and structural consistency, as Figures 10(c), 10(d) and 10(f) show. Figure 10(d) in particular exhibits severe structural deviation from the source images, which

markedly degrades visual fidelity.

In contrast, the complete loss function, combining pixel intensity, gradient and structural losses, enables the model to fully exploit the complementary features of the infrared and visible images. As shown in Figure 10(i), the fusion result obtained with the full loss preserves target saliency, enhances edge clarity and exhibits high visual realism. This confirms that all three terms are indispensable for achieving the best fusion quality.

Table 4. Infrared and visible satellite remote sensing dataset module ablation experiments

Method	SF $\uparrow$	AG $\uparrow$	PSNR $\uparrow$	MSE $\downarrow$	VIF $\uparrow$
w/o STM	7.343	3.545	63.453	0.035	0.536
w/o MARM	9.406	4.127	62.705	0.043	0.516
Proposed	<u>35.367</u>	<u>28.462</u>	<u>64.897</u>	<u>0.025</u>	<u>0.622</u>

Table 5. Infrared and visible aerial remote sensing dataset module ablation experiments

Method	SF $\uparrow$	AG $\uparrow$	PSNR $\uparrow$	MSE $\downarrow$	VIF $\uparrow$
w/o STM	17.447	6.095	62.780	0.041	0.562
w/o MARM	17.291	6.183	62.802	0.040	0.584
Proposed	<u>22.907</u>	<u>9.463</u>	<u>63.715</u>	<u>0.037</u>	<u>0.586</u>

Table 6. Near-infrared and visible satellite remote sensing dataset module ablation experiments

Method	SF $\uparrow$	AG $\uparrow$	PSNR $\uparrow$	MSE $\downarrow$	VIF $\uparrow$
w/o STM	10.992	3.947	69.287	0.018	0.713
w/o MARM	10.957	3.934	69.476	0.019	0.731
Proposed	<u>18.973</u>	<u>8.439</u>	<u>69.862</u>	<u>0.009</u>	<u>0.950</u>

In conclusion, the results from these comparative experiments demonstrate the superior performance of the proposed method. In addition, a series of ablation studies verify the individual contributions of each module within the overall architecture.

5. Conclusion

This paper has proposed an end-to-end fusion network based on decoupled parallel attention and global coupling for infrared and visible remote sensing image fusion. The core idea is to explicitly decouple the fundamentally different contributions of the two modalities: infrared thermal saliency (channel-wise) and visible textural details (spatial), and independently reinforce them before global modeling. The MARM achieves this decoupling through parallel spatial and channel attention branches, while the STM performs global coupling by capturing long-range dependencies at linear compu-

tational cost. An integrated composite loss further guides the network to retain intensity, gradient, and structural information simultaneously. Extensive experiments are conducted on three datasets covering infrared–visible satellite imagery, infrared–visible aerial imagery and near-infrared–visible satellite imagery. Both qualitative and quantitative evaluations demonstrate that the proposed network integrates complementary information from the two modalities more effectively than the compared methods. Notably, it achieves a 38.5% improvement in average gradient over the second-best method on the infrared–visible satellite dataset and a 2.1 dB improvement in PSNR over the second-best method on the near-infrared–visible satellite dataset. Ablation studies validate the complementary roles of the decoupling and global coupling modules, and confirm the necessity of each loss term.

For future work, we plan to extend this framework to incorporate downstream remote sensing tasks such as target detection and semantic segmentation, enabling task-driven optimization along the lines recently explored by task- and semantics-guided fusion methods [42, 43]. We also aim to explore model compression techniques, as well as recent linear-complexity global operators such as selective state space models [37], to reduce the computational footprint of the global coupling stage for potential on-board deployment on airborne or spaceborne platforms.

Acknowledgments

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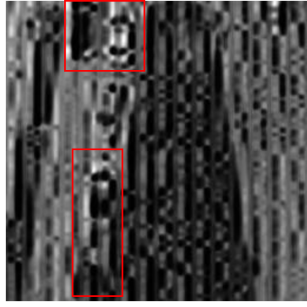
(a) VIS



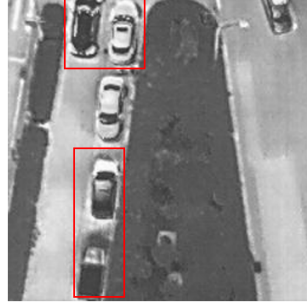
(b) IR



(c) pixel intensity loss only



(d) gradient loss only



(e) structural loss only



(f) pixel intensity and gradient loss



(g) pixel intensity and structural loss



(h) gradient and structural loss



(i) Proposed Loss

Figure 10. Fusion results of loss function ablation experiments